

Mathematical Model of the ReVolt Vessel with Actuators Faults

Stephen Monnet

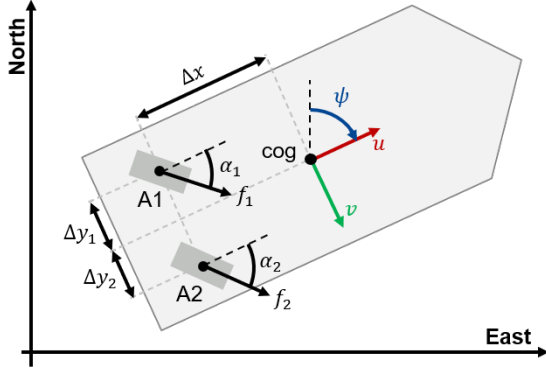


Fig. 1. Configuration of the ReVolt model vessel in sailing mode.

I. INTRODUCTION

This file is quite recent so if you catch any error, please let me know.

II. MODEL

The 3-DOFs dynamics of the ReVolt vessel without faults is

$$\begin{cases} \dot{\eta} = R(\psi)\nu \\ M\dot{\nu} + C(\nu)\nu + D\nu = \tau(\alpha, n) + \tau_{ext} \\ \dot{\alpha} = (\alpha^d - \alpha)/T_\alpha \\ \dot{n} = (n^d - n)/T_n \end{cases} \quad (1)$$

Here, the state vector is defined as $x^T := [\eta^T, \nu^T, \alpha^T, n^T]$ with $\eta := [N, E, \psi]^T$ (north and east positions and heading angle), $\nu := [u, v, r]^T$ (surge and sway speeds and yaw rate), $\alpha := [\alpha_1, \alpha_2]^T$ (azimuth angles) and $n := [n_1, n_2]^T$ (propeller speeds). The control commands $u \in \mathcal{U} \subseteq \mathbb{R}^4$ is defined as the concatenation of α^d and n^d . The generalized force applied by the actuators is τ while τ_{ext} represents exogenous disturbances. The rotation matrix R is defined as

$$R(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

The matrices M , $C(\nu)$, and $D(\nu)$ representing respectively inertia, Coriolis forces and damping are defined as

$$\begin{aligned} M &:= M_{RB} + M_A \\ &= \begin{bmatrix} m & 0 & 0 \\ 0 & m & mx_g \\ 0 & mx_g & I_z \end{bmatrix} + \begin{bmatrix} -X_{\dot{u}} & 0 & 0 \\ 0 & -Y_{\dot{v}} & -Y_{\dot{r}} \\ 0 & -Y_{\dot{r}} & -N_{\dot{v}} \end{bmatrix} \end{aligned} \quad (3)$$

$$\begin{aligned} C(\nu) &:= C_{RB}(\nu) + C_A(\nu) \\ &= \begin{bmatrix} 0 & 0 & -m(x_g r + v) \\ 0 & 0 & mu \\ m(x_g r + v) & -mu & 0 \end{bmatrix} \\ &\quad + \begin{bmatrix} 0 & 0 & Y_{\dot{v}}v + Y_{\dot{r}}r \\ 0 & 0 & -X_{\dot{u}}u \\ -Y_{\dot{v}}v - Y_{\dot{r}}r & X_{\dot{u}}u & 0 \end{bmatrix} \end{aligned} \quad (4)$$

$$D := \begin{bmatrix} -X_u & 0 & 0 \\ 0 & -Y_v & -Y_r \\ 0 & -N_v & -N_r \end{bmatrix} \quad (5)$$

A. Faults

The vessel is equipped with two azimuth thrusters located at the stern. The thrust generated by each thruster is given by

$$\tilde{f}_i = \delta_i f_i = \delta_i \cdot k \cdot n_i \cdot |n_i| \quad (6)$$

where $n_i \in [0, \bar{n}]$ represents the propeller speed of thruster i , and k is the force coefficient. The term f_i is the ideal thrust generated under normal conditions (assuming no faults), and $\delta_i \in [0, 1]$ is the efficiency factor. The actual thrust created by the propeller, $\tilde{f}_i$, accounts for potential loss-of-efficiency faults. When $\delta_i < 1$, the propeller is considered faulty, and $\delta_i = 1$ represents an ideal, fault-free propeller. Additionally, we model failures of each stepper motor controlling the azimuth angle with

$$\dot{\alpha}_i = \epsilon_i \cdot (\alpha_i^d - \alpha_i)/T_\alpha \quad (7)$$

Where $\epsilon_i \in \{0, 1\}$ is the failure flag indicating normal conditions ($\epsilon_i = 1$) or failure ($\epsilon_i = 0$). The generalized force applied by actuator i can be expressed as

$$\tau(\alpha, n) := B(\alpha) \cdot \Delta \cdot K \cdot n^{|2|} \quad (8)$$

where Δ and K are diagonal matrices containing respectively the loss-of-efficiency and force coefficients, $n^{|2|}$ is an abuse of notation representing a vector whose i -th component is $n_i \cdot |n_i|$, and $B(\alpha)$ is a matrix projecting propellers thrust into the vessel's body frame:

$$B(\alpha) := \begin{bmatrix} \cos(\alpha_1) & \sin(\alpha_1) & \Delta y_1 \cos(\alpha_1) - \Delta x \sin(\alpha_1) \\ \cos(\alpha_2) & \sin(\alpha_2) & -\Delta y_2 \cos(\alpha_2) - \Delta x \sin(\alpha_2) \end{bmatrix}^T \quad (9)$$

B. The ReVolt as a Control-Affine System

Based on the presented mathematical model, the vessel can be modeled as a control-affine system, that is

$$\dot{x} = f_{\theta}(x) + g_{\theta}(x)u + w \quad (10)$$

where $\theta := \{\delta, \epsilon\}$ and

$$f_{\theta}(x) := \begin{bmatrix} R(\psi)\nu \\ -M^{-1} [C(\nu)\nu + D\nu - B(\alpha)\Delta K n^{[2]}] \\ -E\alpha/T_{\alpha} \\ -n/T_n \end{bmatrix} \quad (11)$$

$$g_{\theta}(x) := \begin{bmatrix} \mathbf{0}_{3 \times 2} & \mathbf{0}_{3 \times 2} \\ \mathbf{0}_{3 \times 2} & \mathbf{0}_{3 \times 2} \\ E/T_{\alpha} & \mathbf{0}_{2 \times 2} \\ \mathbf{0}_{2 \times 2} & I_{2 \times 2}/T_n \end{bmatrix} \quad (12)$$

where $E \in \{0, 1\}^{2 \times 2}$ is a diagonal matrix containing the azimuth failure flags ϵ_i . Finally, the exogenous disturbances are modeled as

$$w := \begin{bmatrix} \mathbf{0}_{3 \times 1} \\ \tau_{ext} \\ w_{\alpha} \\ w_n \end{bmatrix} \quad (13)$$

C. Sensor Faults

According to [1], the most common sensor faults can be classified as either

- Bias
- Drift (increasing bias)
- Scaling (bad calibration)
- Dropout (we lose one instance of data)

REFERENCES

- [1] E. Balaban, A. Saxena, P. Bansal, K. F. Goebel, and S. Curran, "Modeling, detection, and disambiguation of sensor faults for aerospace applications," *IEEE Sensors Journal*, vol. 9, no. 12, pp. 1907–1917, 2009.